

4C11 Coursework 3

5630C

April 7, 2025

Abstract

A Recurrent Neural Network (RNN) based on MLP is trained in a supervised manner to learn the macroscopic constitutive time-dependent relationship between strain and stress in a viscoelastic material. For this 1D case, it is shown there need not be more than 1 memory parameter. Changes in MLP architecture show little effect on performance.

Problem Definition

A neural network is to be trained to predict the macroscopic stress σ_t time series given a macroscopic strain ε_t time series.

The mapping defining the input and output of the macroscopic constitutive model to be learned with T time steps is

$$\psi_\theta =: \{\varepsilon_i\}_{i=0}^T \mapsto \{\sigma_i\}_{i=0}^T \quad (1)$$

The problem is memory-dependent, meaning σ_t depends on the strain history $\{\varepsilon_i\}_{i=0}^t$. The analytic stress history, used to calculate the current stress, and its gradient are defined as

$$\hat{\zeta}(t) = \int_0^t \varepsilon(\tau) \exp[-\beta(t - \tau)] d\tau \quad (2)$$

$$\frac{d\hat{\zeta}}{dt}(t) = -\beta\hat{\zeta}(t) + \varepsilon(t) \quad (3)$$

Method

In order to account for the memory dependence, a Recurrent Neural Network (RNN) was trained to learn the data. This consists of two Multilayer Perceptrons (MLPs) g and f (with trained parameters θ_g and θ_f) to learn the mappings

$$g := \{\varepsilon(t), \hat{\zeta}(t), \theta_g\} \mapsto \frac{d\hat{\zeta}}{dt}(t) \quad (4)$$

$$f := \{\varepsilon(t), \hat{\zeta}(t), \theta_f\} \mapsto \sigma(t) \quad (5)$$

With the RNN memory state update rule

$$\tilde{\zeta}_{t+1} = \tilde{\zeta}_t + \delta t \frac{d\hat{\zeta}}{dt}(t) \quad (6)$$

Where δt is the time difference per iteration.

This way the entire stress history for T steps can be reconstructed by iteratively evaluating the RNN T times using the strain history as input.

For a 1D stress/strain the memory gradient $\frac{d\hat{\zeta}}{dt}(t)$ (and the memory state itself) is a single value for any one time t . Thus, the hypothesis was that only 1

memory (hidden/recurrent) parameter is necessary for the network. The performance was compared with the same RNN with 2 and 100 hidden parameters to confirm this.

The hidden layer structure of the MLPs f and g was tuned. The networks were tested with 1. one hidden layer with 500 neurons, 2. three hidden layers with 100 neurons each, 3. three hidden layers with 500 neurons each. f and g both took on this structure.

The loss function was mean square error (MSE) and the optimiser used was AdamW. The AdamW learning rate and weight decay hyperparameters were tuned.

Results and Discussion

Using more than 1 memory parameter seems to have minimal effect on ultimate performance, shown in fig. 1. While it should be possible for a network with 1 memory parameter to perfectly predict the output, increasing the number of memory parameters may make it slightly easier for the network to learn the relationship.

The hidden layer structure seemed to have little impact on performance, as shown in fig. 2.

A learning rate of 0.001 was found optimal for all tests. No disconnect between training loss and test loss was encountered so weight decay was always zero.

An comparison between the true and predicted stress history for an example from the test set is shown in fig. 3. The RNN performs well.

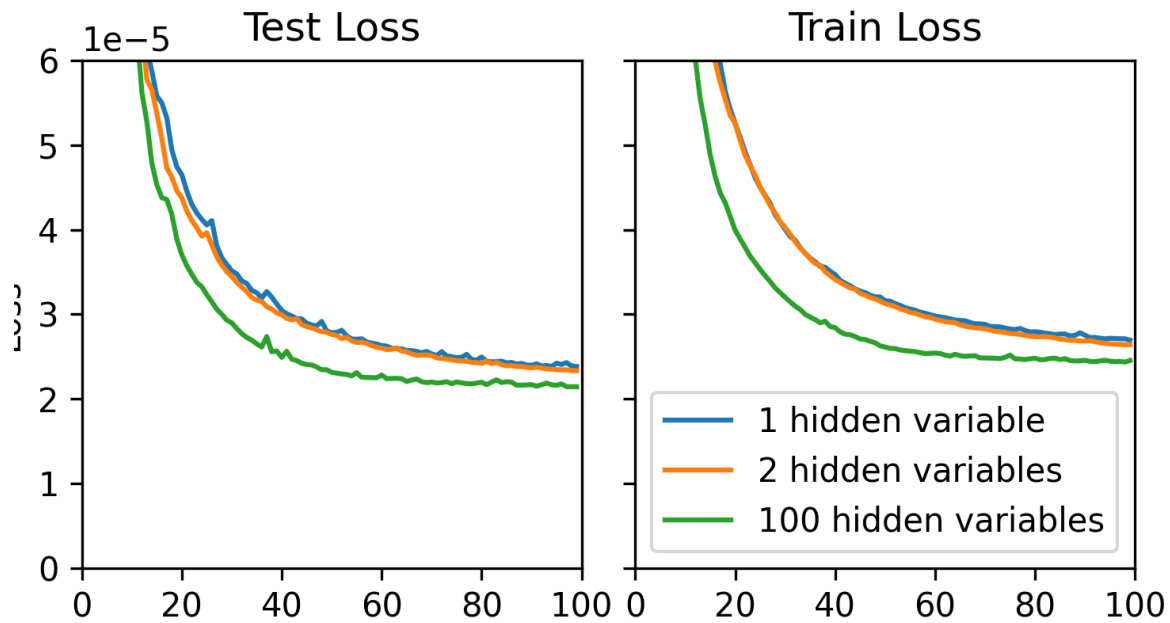


Figure 1: Training and test loss performance for varying memory parameter count using normalised stress/strain. While increasing the number of hidden parameters slightly improves the final loss, all the cases seem to be on a path to converge to the same loss. All 3 tests were conducted with 3 hidden layers with 100 neurons each.

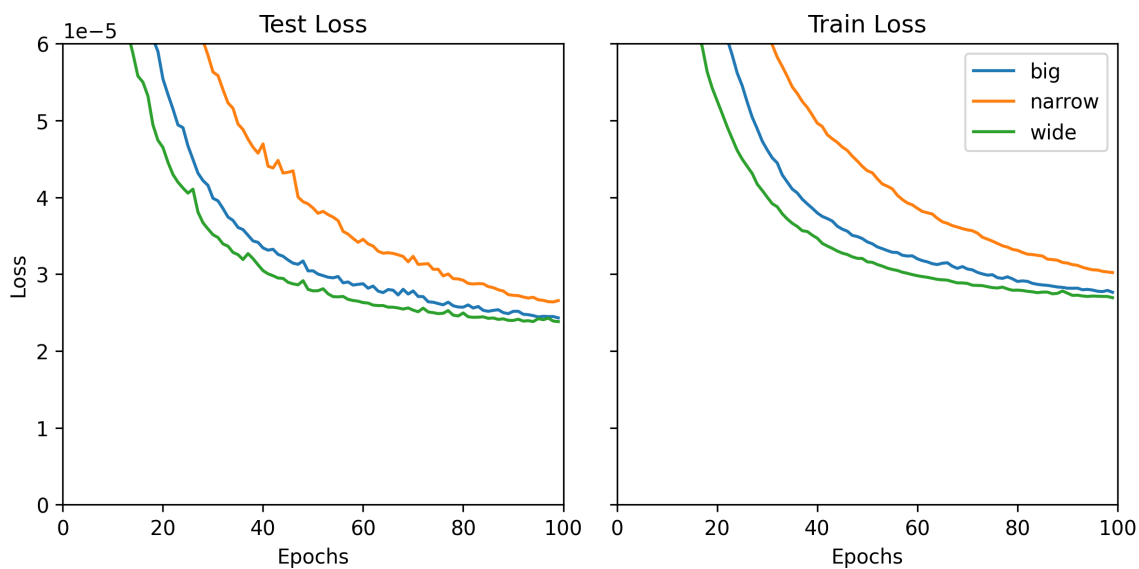


Figure 2: Training and test loss performance for varying hidden layer structure using normalised stress/strain. The structure used seemed to have little effect on the final loss, and all the cases seem to be on a path to converge to the same loss.

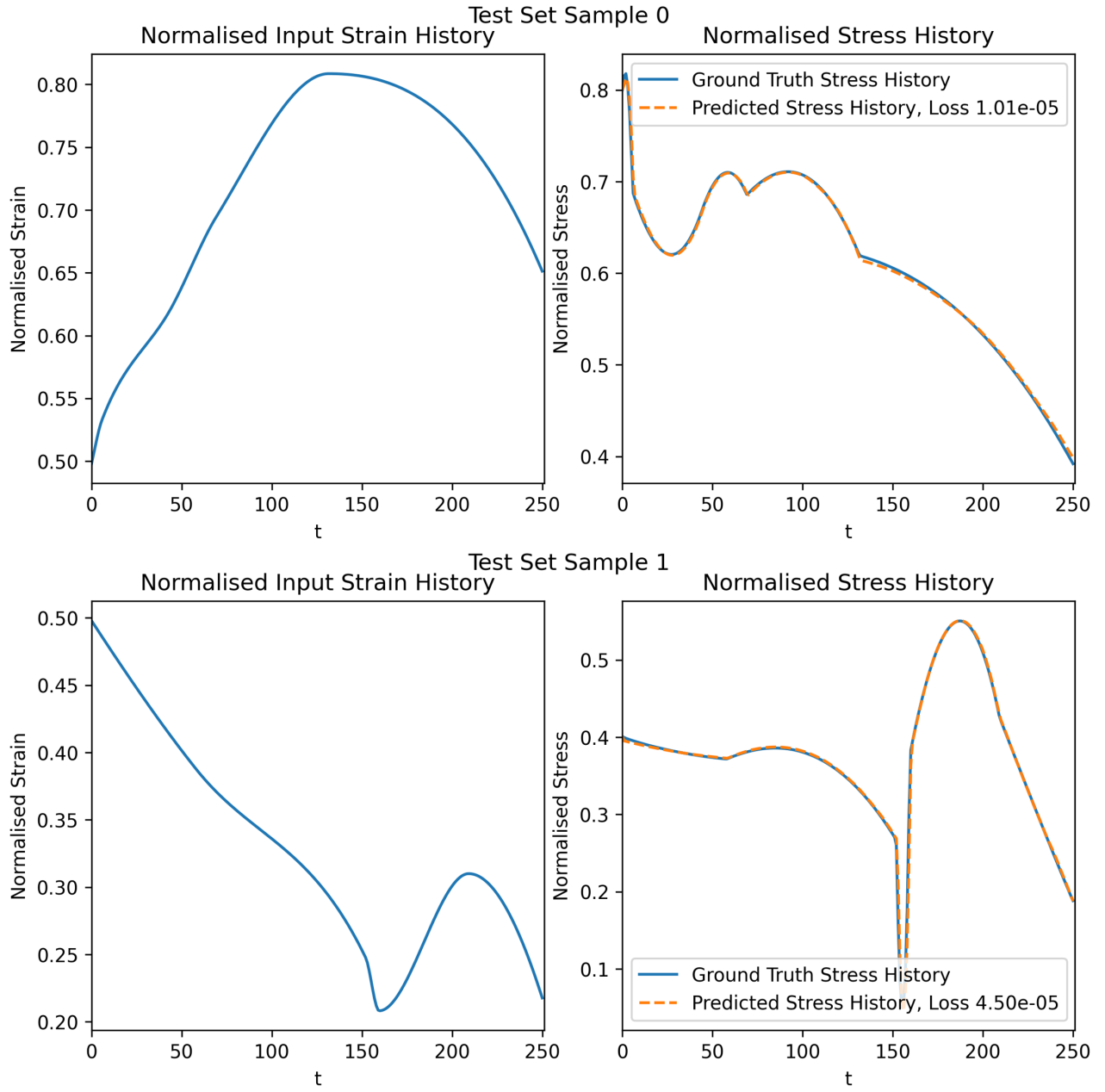


Figure 3: Two examples of the RNN's stress history predictions from the test set, given the strain history.